

Sparkdrift GP: Design Package

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Sparkdrift GP: Executive Summary

1. **What:** an original, premium-feel arcade kart racer for phones, in the spirit of a top console kart racer but with its own world, characters and music.
2. **Scale:** 100 original racers in 10 themed sets of 10, 42 kart bodies, 18 wheel styles in 5 sizes, 12 spoilers, 10 exhausts, 8 bumpers, 32 paints × 5 finishes, and 20 tracks (4 cups of 4 plus 4 bonus).
3. **Feel:** drift for a 3-tier boost, a timed start boost, drafting, bump rules by weight class, light rubber-banding, and 8 skill-based items that each have a counter.
4. **Modes:** Grand Prix (12-racer field), Time Trial with ghosts, Versus with 6-character code rooms for 2 to 4 humans (CPU fill to 12), optional Battle, Daily Challenge with a leaderboard.
5. **Economy:** coins from racing buy everything; money buys cosmetics only; every kart body shares one stat budget (no pay-to-win).
6. **Multiplayer:** unambiguous 20-symbol code alphabet, host-authoritative items with lag compensation, client prediction, host migration, CPU takeover on drop.
7. **Sound first:** class-specific layered engines, tiered drift/boost sounds, an announcer, character barks, per-track ambience, and a six-stem dynamic score that reacts to position and the final lap, through a ducking mixer and master limiter.
8. **Vertical slice:** 8 characters, 6 karts, full garage, 4 tracks, 4-player code lobbies, all items, 3-tier drift boost. A playable browser build ships alongside these docs.
9. **Production:** 24 months, ~85 people, \$10M; Unity 6 + Nakama + FMOD recommended; risks and the performance budget are in 06 .
10. **Docs:** 01 game design, 02 roster, 03 karts/mods, 04 tracks, 05 vertical slice, 06 production/risks/perf. All names and stats are generated from one data source and verified by script.

Sparkdrift GP: Game Design Document

Version 1.0 · Original IP · Feel reference only: "a polished arcade kart racer in the spirit of a premium console kart racer". No third-party characters, tracks, names or audio are used anywhere in this project.

1. Vision and pillars

One-line pitch. A saturated, readable, fast arcade kart racer for phones: 100 expressive racers, 40+ customisable karts, 20 tracks with shortcuts, skill-based items, 1 to 4 humans in code rooms, and a score that reacts to every overtake.

Design pillars

- Readable at speed.** Every kart, hazard and item reads from silhouette and colour at thumbnail size. The racing line is always the brightest, cleanest shape on screen. The HUD never covers the racing line.
- Drift is the instrument.** Drifting is the main skill expression: choose a line, charge a tier, release at the right moment. Three tiers, three distinct audio-visual signatures.
- Skill beats luck.** Items can be held as shields, dodged, baited and countered. Rubber-banding is light enough that a better driver can still pull away.
- Customisation is identity, not power creep.** Every kart body has the same power budget (stat sum 22). Mods trade one stat for another, capped at ± 1.5 per stat. No pay-to-win: money buys cosmetics only; coins buy everything else, earned by racing.
- Sound is half the game.** Engines, drift squeals, boost whooshes, an announcer, crowd, barks and a position-reactive score are designed as one mix, not as an afterthought (see section 14).
- Friends in ten seconds.** A host taps once, gets a 6-character code, says it aloud, and friends join in seconds. If someone drops, a CPU takes over.

Target feel

Metric	Target
Frame rate	60 fps on flagship phones, stable 30 fps on mid-range
Input latency (touch to steering)	below 60 ms
First load	under 8 s on a good phone (core download under 6 MB for the web slice)
Race length	3 laps, about 2:30 to 4:30
Session loop	a race, a reward screen, a garage tweak, the next race in under 90 s

2. Platforms and technology targets

- **Production target:** iOS and Android phones and tablets (ARM64), Bluetooth gamepad support.
- **This repository's playable slice:** a free static browser build (three.js, WebAudio, WebRTC) that runs on iPhone Safari and Android Chrome. It implements the vertical slice defined in `05-vertical-slice-plan.md` with the same numbers as this document.
- Units: metres, seconds. HUD speed readout = $m/s \times 4$.

3. Controls

3.1 Touch (default)

Control	Position / behaviour
Steering slider	A horizontal slider, bottom-left thumb zone, 38% of screen width. Dead zone 6%, response curve $y = \text{sign}(x) \cdot x ^{1.25}$, optional auto-centre.
Accelerate	Automatic by default (Assist: Auto-Accel on). Hold-to-accelerate is available for experts.
Brake / reverse	Pedal under the right thumb, lower centre-right.
Drift / hop button	Large round button, bottom-right, 14% of screen height. Tap to hop; hold while steering to drift. Release to fire the boost.
Item button	Right side, above the drift button. Tap to use, hold to keep the item trailing behind as a shield, swipe up/down to aim forward/back.
Look-back	Small button near the item slot. Hold to view behind.
Pause	Top-right corner, away from controls.

Optional **tilt steering** (gyro) with calibration, 12° dead zone, sensitivity slider. Touch controls can be moved and resized in settings. Left-handed layout mirrors everything.

3.2 Gamepad (Bluetooth)

Standard mapping: left stick steer, right trigger accelerate, left trigger brake, **A** drift/hop, **X** item, **Y** look-back, **Start** pause, d-pad menu navigation.

3.3 Accessibility

Auto-accelerate, steering assist (adds a soft lane pull of up to 20%), colour-blind-safe item and drift-tier indicators (shape + colour), reduced camera shake, reduced flashing, large HUD, haptics slider, subtitles for the announcer and barks, one-handed mode (steer by tilt and drift by screen tap).

4. Physics and handling targets

All numbers are tuned for 60 Hz fixed-step simulation (30 Hz render fallback runs the same fixed step with interpolation).

4.1 Stats and how they map to physics

Every kart has five stats (1 to 10): **S** top speed, **A** acceleration, **H** handling, **G** drift grip, **W** weight. Final stats = body + character-class modifier + wheel + spoiler + exhaust + bumper (each clamped so total mod per stat is within ± 1.5), then clamped to 1 to 10.

Stat	Formula	Range
Top speed	$v_{\max} = 34 + 1.2 \cdot S \text{ m/s}$	35.2 to 46.0 m/s (141 to 184 on the HUD)
Acceleration	time to 90% of $v_{\max}$ on tarmac $t_{90} = 6.4 - 0.38 \cdot A \text{ s}$; $a(v) = a_0 \cdot (1 - (v/v_{\max})^2)$, $a_0 = 1.472 \cdot v_{\max} / t_{90}$	6.0 s down to 2.6 s
Handling	yaw-rate cap $105^\circ/\text{s}$ scaled by $(0.72 + 0.03 \cdot H)$, limited by lateral acceleration	wider turns at higher speed
Drift grip	lateral acceleration cap $18 + 1.6 \cdot G \text{ m/s}^2$	19.6 to 34 m/s^2
Weight	mass $0.7 + 0.12 \cdot W$	0.82 to 1.9

Other base values: brake 30 m/s^2 , coasting drag 3.0 m/s^2 , reverse top speed 10 m/s , off-road top speed $\times 0.55$ and acceleration $\times 0.6$, rough shortcut ground $\times 0.8$ top speed, boost pad 1.1 s at $\times 1.28$.

4.2 Character classes

Classes add small modifiers to the body stats:

Class	S	A	H	G	W	Role
Light	-0.5	+1.0	+0.5	0.0	-2.0	Fast acceleration, quick recovery, weak in bumps (loses 15% speed on a heavy hit and is spun out above 14 m/s impact)
Medium	0	0	0	0	0	Balanced
Heavy	+0.8	-1.0	-0.5	+0.2	+2.5	Slow acceleration, high top speed, wins bump contests (loses only 3% speed)

4.3 Steering model

Heading and velocity are separate. Steering rotates the heading at yaw rate ω , capped by $\omega \leq a_{\text{lat}} / v$ so cornering is bounded by grip. Lateral velocity (slip) decays at 11 s^{-1} on a normal line (feels glued), 1.6 s^{-1} while drifting (feels sliding), and 7.0 s^{-1} off-road.

4.4 Surfaces

Surface	Top speed	Accel	Grip	Notes
Road	$\times 1.00$	$\times 1.00$	normal	racing line
Rough (shortcut ground)	$\times 0.8$	$\times 0.85$	slightly loose	cut risks speed loss without a boost
Off-road (grass, sand, snow)	$\times 0.55$	$\times 0.6$	low	slows to a crawl, camera shakes
Ice	$\times 1.00$	$\times 0.9$	very low	reduced by Studded wheels
Boost pad	$1.1 \text{ s} \times 1.28$	boost	n/a	arrows glow

Surface	Top speed	Accel	Grip	Notes
Walls	n/a	n/a	n/a	see bump rules

5. Drifting and boost

1. **Hop.** Tap the drift button above 17 m/s to hop (small jump, 0.9 m/s upward impulse for feel only). Hold steer left or right to commit to a direction.
2. **Drift.** While held, the kart slides at up to **32°** between heading and velocity. Steering *into* the drift multiplies yaw by $\times 1.25$ (tightens), steering *against* multiplies by $\times 0.72$ (widens). Steer commitment sets the charge rate: full steer charges at $\times 1.0$, no steer at $\times 0.55$. The neutral drift yaw rate is **0.62** of the kart's cornering cap ($\text{baseYaw} \times \min(1.2 \cdot \text{yawMax}, 1.35 \cdot a_{\text{lat}}/v)$), so a drift always carves a slightly wider arc than a normal turn until the player steers into it.
3. **Charge tiers (seconds of drifting at full commitment):** Blue / Orange / Purple = **0.9 / 1.9 / 3.1 s**.
4. **Release.** Releasing the button fires the boost: **0.8 / 1.5 / 2.3 s** at $\times 1.14 / 1.22 / 1.3$ top speed for tiers 1 to 3.
5. **Feedback.** Spark colours: blue, orange, purple. Sparks change at each threshold with a rising "tick" tone; tier 3 adds a screen-edge streak and a bass swell. Releasing fires the tiered whoosh and a camera FOV kick.
6. **Start boost.** On the countdown, press accelerate within **0.28 s** of "GO": **1.3 s at $\times 1.25$** . Pressing too early stalls the engine for 1.0 s (sputter sound, smoke).
7. **Stacking.** Boosts do not stack: the strongest multiplier applies and the longest remaining time is kept.

6. Bumping, drafting and slipstream

6.1 Kart-to-kart bumps

- Collision is circle vs circle, radius **1.15 m**, restitution **0.45**.
- Mass $m = 0.7 + 0.12 \cdot W$. The impulse is split by inverse mass, so a heavy kart pushes a light kart further.
- Speed loss on a heavy-vs-light hit: light loses **15%**, heavy loses **3%** of forward speed.
- Relative impact speed above **14 m/s** spins the lighter kart for **0.6 s** (hit-stun, cannot steer; still has momentum). Stars and ghosts ignore this.
- Side contact during a drift transfers no charge but gives the attacker a short +4% speed shove for 0.4 s ("rub is racing").

6.2 Walls

Glancing contact (angle below 25°) scrapes for **8%** speed loss per second of contact with sparks. Head-on contact (angle above 60°) cuts speed by **50%** with a screen shake. Walls never dead-stop a kart: a minimum outward push keeps it moving.

6.3 Draft and slipstream

- A kart is in a draft when it is **4.0 to 22 m** behind another and within **1.8 m** laterally of its path.
- While in draft: **+6%** top speed (×1.06) and +10% acceleration.
- After **2.0 s** of continuous draft, leaving the wake triggers a **slingshot**: ×1.12 for 1.2 s. The HUD draws a curved air-line behind the lead kart.
- Draft charge starts after 1.2 s inside the zone and decays at twice the build rate when you leave it.

7. Items

Eight items, all with original names. Items are collected from **item rows** (rainbow orbs, row of 4 to 6 across the road). Each racer holds **one item at a time** (a second item is only picked up when the slot is empty; a shield item counts as held).

#	Item (original name)	Based on	Exact behaviour	Counterplay
1	Turbo Pod	boost mushroom	Instant boost: 1.8 s at ×1.35 top speed, also removes 50% of off-road slowdown for the duration. Rarely a Pod Trio : 3 uses, 1.4 s at ×1.3 each.	Use on a straight or off-road shortcut; combine with a drift boost for max speed; the boost can be wasted if used before a wall
2	Rebound Disc	shell	Fired forward (tap) or backward (swipe down) at 58 m/s , bounces off walls up to 4 times, lifetime 8 s. Holding the button trails it behind you as a shield that absorbs one hit (it pops with a sound). A hit spins the victim for 1.2 s and removes 60% of speed.	Shield, Phantom Veil, Nova Core, steering around its bounce prediction line (a faint trail), out-driving it
3	Peel Trap	banana	Drop behind you, or throw forward 12 m. A kart that touches it spins for 0.9 s and loses 25% of speed. Stays until hit; each racer can have at most 3 on the course. Holding it trails behind as a shield.	Shield, steer around it, burn it with a Rebound Disc
4	Storm Jolt	lightning	Only obtainable from rank 6 or lower. After a 1.5 s thunder arming (a visible cloud above the leader and an audible rumble), every racer <i>ahead</i> of you shrinks for 6.0 s at ×0.78 top speed and drops a shield item.	Be protected by Nova Core or Phantom Veil before it fires; drive in the draft to close the gap; not obtainable by leaders
5	Nova Core	star	Invincible for 7.0 s at ×1.2 top speed, ignores off-road, spins any kart it touches (as a Disc hit), crushes hazards. The last 1.5 s flash to warn.	Dodge, don't collide, let it pass
6	Phantom Veil	ghost	5.0 s of intangibility: passes through karts and items; immune to Disc, Peel, Spill, Rocket and Storm Jolt. On activation it steals the held item (not shields) from the nearest racer within 25 m ahead. Cannot use items while ghosted.	Keep a shield item up (shields cannot be stolen); stay out of 25 m ahead of a Veil holder
7	Slick Spill	oil	Drop behind: a puddle radius 1.8 m lasts 15 s. A kart crossing loses steering for 1.2 s and 10% speed.	Steer around it, Trail or Mud wheels halve the slip time, Nova Core
8	Hornet Rocket	rocket	Only obtainable from rank 3 or lower. A homing rocket at 72 m/s seeking the race leader along the racing line. The target gets a lock warning for 2.0 s (beeps accelerating, red reticle on the HUD). A hit spins out 1.6 s.	Perfect Brace : press the item button in the 0.25 s window as the rocket arrives to burn your held item and cancel it; also Phantom Veil, Nova Core, or hiding behind another kart in a narrow section

7.1 Item distribution (percent chance, by rank in a 12-kart field)

Item	1st	2nd to 4th	5th to 8th	9th to 12th
Turbo Pod	18	22	22	18
Pod Trio	0	4	10	14
Rebound Disc	30	22	14	8
Peel Trap	30	20	10	4
Slick Spill	16	12	8	4
Phantom Veil	6	8	10	10
Nova Core	0	4	12	20
Storm Jolt	0	0	6	12
Hornet Rocket	0	8	8	10

Items are blocked for the first **8 s** of a race (only Pod, Disc, Peel and Spill can drop) and no item row exists in the first corner or within 120 m of the grid.

7.2 Defensive play

- A held **shield** (Disc, Peel, Spill used as trailing items) blocks one hit of any kind but a Storm Jolt.
- **Perfect Brace** (see Rocket) turns any held item into a counter.
- Weight class and wheel compound change how much damage an item does: Heavy karts lose 20% less speed from Peel/Spill; Light karts recover from spin-outs 0.2 s faster.
- Item boxes respawn after 5 s. A skilled leader holding a shield can defend the lead without spending it.

8. Rubber-banding and CPU behaviour

- **Light rubber-banding.** CPU karts multiply their top speed by `0.97` plus a gap term: from 0.965 to 1.035 depending on the distance to the *player* (gap normalised over ± 90 m). Players are never speed-boosted by rubber-banding. The comeback comes from item distribution only.
- Skilled players can still pull away: a perfect line, three-tier drifts and no hits give about +6% over a CPU's average speed.
- **CPU skill tiers:** Easy (aggression 0.3, 15% mistakes), Normal (0.5, 8%), Hard (0.7, 4%), Expert (0.9, 1.5%). Mistakes mean late braking, wide lines or missed drifts.
- **CPU behaviours.** They follow the racing line with per-driver lane offsets, drift on long corners, take shortcuts according to skill and rank, use items sensibly (fire backwards at close followers, hold shields, drop traps in narrows), and do not catch up unfairly after player spin-outs.
- **CPU field.** Grand Prix: 12 racers (you plus 11 CPUs). Versus: humans (2 to 4) plus CPUs filled to 12.

9. Game modes

9.1 Grand Prix

Four cups of four tracks each: **Seedling Cup, Copper Cup, Tempest Cup, Zenith Cup**. 12-racer field. Points: 15, 12, 10, 8, 7, 6, 5, 4, 3, 2, 1, 0 per race. Trophy at the end: Gold for first overall, Silver second, Bronze third. Engine classes: 100cc, 150cc, 200cc (speed $\times 0.93$, $\times 1.00$, $\times 1.07$) and Mirror mode.

9.2 Time Trial

Solo. Track-specific personal best and ghost. Developer ghosts at Bronze, Silver and Gold times. Ghost data is compact: input stream sampled at 30 Hz with quantised steering (6 bits), drift/item flags, and a position checkpoint every 2 s for error correction, around 30 bytes per second, around 8 KB per 3-lap run after delta compression and DEFLATE. Up to 3 ghosts at once (PB, friend, developer).

9.3 Versus (code rooms)

Private rooms for 2 to 4 humans (CPU fill to 12). The host picks the track, laps (1 to 5), items on/off, CPU fill and CPU difficulty. Quick Match (public matchmaking) uses the same netcode.

9.4 Battle (optional, post-launch if it does not delay the racer)

Three balloons per racer, 4-player arena maps, items only, 3-minute timer. Specified in the roadmap; not part of the vertical slice.

9.5 Daily Challenge

One seeded race a day (track, items and CPU seed fixed for everyone). One attempt per day counts for the global and friends leaderboards; unlimited practice. Rewards: 200 coins for finishing, +25 coins per day of streak up to 7 days, and a rotating cosmetic at 7 days.

9.6 Laps

Default **3**; host can set **1 to 5**.

10. Progression and economy

10.1 Earnings

Source	Coins
Finish position payout (1st to 12th)	120, 100, 85, 70, 60, 50, 42, 35, 28, 22, 16, 10
On-track coins	1 each, up to 40 per lap
First race of the day	+100
Grand Prix trophy	Gold 400, Silver 250, Bronze 150
Daily challenge	200 + streak (25 × day, max 7)
Achievements	50 to 500 each

Typical average: about **190 coins per race** (4 to 5 minutes including menus, so about 45 coins per minute).

10.2 Prices

Category	Total coins to buy everything
Characters (coin-bought: 61 of 100)	75,180
Kart bodies (42)	34,050
Wheel styles (sizes and rim colours free)	8,100
Spoilers	4,150
Exhausts	3,400
Bumpers	2,450
Paint, finishes, two-tones, decals	6,320
Everything	133,650

Unlock plan: most content is bought with coins; the rest is earned from trophies and achievements. A dedicated set of 12 free starters is available on first launch. Players can unlock all 100 characters in about **28 hours** of play, and everything else in roughly 50 hours; none of it requires money.

10.3 No pay-to-win

- Money (optional) buys **cosmetics only**: kart skins, trails, emotes, announcer packs, horn sounds, profile frames. These have no stats and are marked "Cosmetic".
- Every body has the same stat sum (22). Mods have trade-offs; total modifier per stat is capped at ± 1.5 .
- Online matches show **no stat badges**; karts are matched by cup class, not wallet.

10.4 Save and cloud profile

Local save in IndexedDB/SQLite with a version number, mirrored to a cloud profile tied to the player account (platform sign-in or email). Conflict rule: per-record last-write-wins with max() for currencies spent only counted once, via a transaction log.

11. Multiplayer flow and netcode

11.1 Room codes

- **Alphabet (20 unambiguous symbols):** A C D E F H J K M N P R T U W X Y 3 4 7 . Removed look-alike and sound-alike symbols: 0 0 Q , 1 I L , 5 S , 2 Z , 8 B , 6 G , 9 , V . Codes are shown in two groups of three, e.g. K7A MPC , and read aloud with NATO-style words.
- Space: $20^6 = 64,000,000$ codes. A code expires after 2 hours of inactivity; the server checks collisions.
- Rooms are stored on a simple matchmaking service (a serverless key-value store with a TTL; WebSocket signalling).

11.2 Flow

1. **Host** taps *Create Room* → server returns a code → lobby opens with host settings (track, laps, items on/off, CPU fill, difficulty).
2. **Guests** enter the code → server returns the host's connection offer → peers connect (WebRTC data channels, or dedicated relay).
3. **Lobby** shows each player (avatar, kart, ready tick, **connection status**: green/amber/red bars with ping in ms, "relayed" tag).
4. Host starts → synchronised 3-2-1 countdown (all clients start on a shared server time).
5. Race → results → return to lobby.

11.3 Join, leave, drop

- **Join late:** blocked during a race; allowed in the lobby until 4 humans.
- **Leave:** a CPU takes the kart immediately (same position, speed, item).
- **Drop (connection lost):** the other players see a "reconnecting" ring; the kart is held by a CPU for 6 s; if the player returns within 30 s they take it back; otherwise the CPU keeps it.
- **Host migration:** every client keeps a replica of the lobby/race state. If the host drops, the client with the lowest RTT to the others becomes host (tie-break: lowest peer id). The new host announces the migration, race state is re-sent in one snapshot, and the countdown "pauses briefly" (max 3 s) to resume.

11.4 Netcode model

- **Own kart:** client-side prediction. Movement is simulated locally and is authoritative for its own position.
- **Remote karts:** state snapshots at **20 Hz** (position quantised to 1 cm, heading 8 bit, speed, drift state, boost flags, animation bits) and **interpolated 100 ms** in the past, with dead-reckoning (extrapolation) up to 250 ms.

- **Items:** host-authoritative events with timestamps. Spawn events carry the host's time; the clients **fast-forward** the projectile by RTT/2 so it appears where it should. Hit resolution is decided on the *victim's* client (victim sees fair dodges) and validated by the host against a 120 ms rewind window; late hits are accepted only inside the window.
- **Race results:** host-authoritative; checkpoints validated by sequence number.
- **Bandwidth:** about 20 players × 50 bytes × 20 Hz = well under 25 KB/s for 4 humans + CPU on host.
- **CPUs:** simulated by the host, streamed as normal karts.
- **Anti-cheat:** speed envelope check, checkpoint order, item-spawn rate check; no stat badges to avoid targeting.

11.5 Connection status UI

A coloured bar icon in the lobby and in-race corner: green (<60 ms), amber (60 to 140 ms), red (>140 ms or loss >5%), grey (reconnecting). Tapping shows RTT, loss, relayed/direct.

12. HUD

Element	Placement	Notes
Race position ("3rd / 12")	top-left	big numerals with a colour pop on change
Lap counter ("Lap 2 / 3")	top-left under position	flash on new lap
Item slot	top-centre, offset	ring shows shield state; rocket warning replaces the reticle
Minimap	top-right	rotated track outline, coloured dots, player arrow
Speed readout	bottom-centre small	m/s × 4
Drift charge arc	around the kart (3D sparks) and thin bar under the drift button	colour+shape coded
Race timer / lap split	top-right under the minimap	
Boost / draft indicators	around the screen edge	subtle
Controls	bottom corners	semi-transparent, never over the road centre
Notifications	upper centre, short	"Shortcut!", "Perfect start!", "Final lap!"

The racing line (the central 40% of the screen width from the kart to the horizon) is kept free of HUD elements.

13. Art direction

- **Style:** stylised PBR, "high-end stylised console kart". Saturated but controlled: each track palette has a dominant hue, an accent hue and a neutral. No neon sludge, no uncanny realism.

- **Materials:** glossy clear-coat paint with flake, matte rubber with tread normal map, brushed metal rims, cloth on characters, subsurface-looking fur.
 - **Lighting:** one warm key light, a cool fill, and a strong rim light on karts so they separate from the background at speed. Baked ground lightmaps per track; soft cascaded shadows only on the player and nearest rivals.
 - **Silhouette rules:** each character has a unique head and body silhouette (see `02-roster.md`); each kart family has its own shape language (Cruiser: friendly arcs, Dart: spears, Drifter: low wedges, Bruiser: boxes, Rocket: bottles/flames, Oddball: props).
 - **Effects:** boost trails (stretching ribbons), drift sparks (blue/orange/purple), dust and snow kick-up per surface, screen shake on big hits (max 6 px, 0.25 s), speed lines at boost, FOV kick +8° on boost.
 - **Readability:** hazards use a fixed visual language (yellow-black chevrons, pulsing telegraph decal), shortcut entries use arrows and a light-gate.
 - **Characters** are expressive and thumbnail-readable: eyes and brows carry emotion, 3 expression states (neutral, joy, hit).
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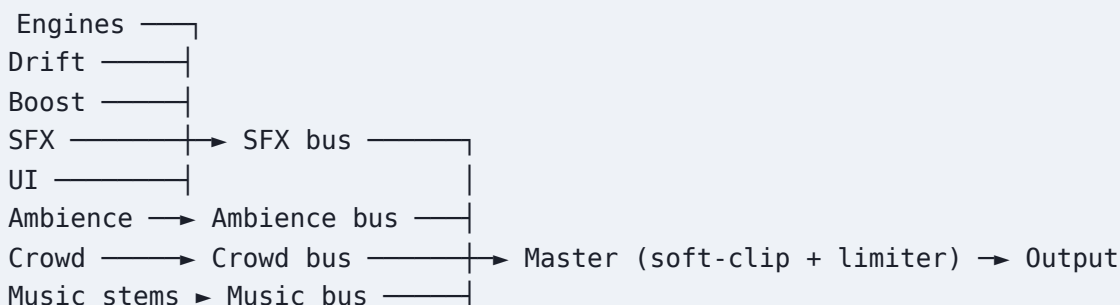
14. Audio and sound design

Sound is the top priority of this project. The audio system is designed as an integrated, dynamic mix.

14.1 Pillars

1. **Everything the player does is audible and satisfying:** engine, drift, boost, hop, land, bump, item.
2. **Information first:** the mix lets the player hear a rocket lock, a thunder arming, a shortcut gate or a rival's engine behind them.
3. **Dynamic music:** layered stems that react to position, speed, items, and the final lap.
4. **Never harsh:** a master limiter, ducking, and equal-loudness mastering (target -16 LUFS, true peak -1 dB).

14.2 Mixer topology



Announcer — Barks — Voice bus — (Voice ducks Music -8 dB, Ambience -4 dB, 0.15 s attack / 0.6 s release)

- Master: gentle soft-clipper then a lookahead limiter (threshold -1.5 dBFS, ratio 20:1, 3 ms attack, 120 ms release).
- Reverb sends per environment: Open (meadow, savanna), Tunnel (cave, tube), Canyon, Hall (foundry, spaceport), Metal (harbor), Ice.
- Ducking: voice → music/ambience; crash → engines (-6 dB, 150 ms); rocket warning → music (-5 dB).
- Mobile: max 32 simultaneous voices, priority by event class, with voice stealing for low-priority sounds. Spatial audio for the three nearest rivals (equal-power panning, distance attenuation, Doppler via playback rate).

14.3 Engines (layered, per class)

Each class has its own engine character:

Class	Character	Base fundamental (idle to redline)	Layers
Light	Buzzy two-stroke / small electric whine	55 to 210 Hz	body saw, high harmonics, intake hiss, whine
Medium	Tuned four-cylinder	40 to 160 Hz	body, mid growl, intake, turbo spool
Heavy	Deep V8 rumble	28 to 110 Hz	sub, body, burble, exhaust crackle

Each class ships **six RPM layers** (idle, 25%, 45%, 65%, 85%, redline) rendered as seamless loops; the game crossfades adjacent layers by RPM and bends pitch $\pm 12\%$ inside each. RPM follows speed with gear-like steps for flavour (a shift dip on the way up). Layers beyond the engine: turbo spool (rises with throttle, whoosh on lift), backfire pops on release, boost roar, off-road rumble and tyre noise by surface.

14.4 Drift, tyres and wind

- **Drift squeal:** a resonant noise loop whose pitch and gain follow slip angle and speed. Different timbres for road, snow and sand (road squeal, snow crunch, sand hiss).
- **Tier ticks:** blue/orange/purple threshold tones rising in pitch, plus a spark crackle layer.
- **Wind:** speed-driven broadband noise, widening with FOV kick.

14.5 Boosts, items and impacts

- **Boost whooshes by tier** (tier 1 short airy, tier 2 wider with a rising whistle, tier 3 deep with a sub hit and shimmer), start boost "punch", Turbo Pod "pop-whoosh", boost pad "zip".
- **Item pickup** (sparkly arpeggio), roulette tick while the slot spins, **use** and **hit** sounds for every item (disc launch/bounce/pop, peel splat, spill glug, jolt thunder with 1.5 s arming rumble, nova jingle loop, veil whisper, rocket launch/flight loop/lock beeps/explosion).

- **Impacts:** kart bump (clunk, weight-scaled), wall scrape (metal on stone), head-on crash (thud + shake whump), spin-out (cartoon whirl).

14.6 Announcer and barks

- **Announcer:** an energetic voice: "3, 2, 1, GO!", "Final lap!", "Lap two", position callouts ("You're in 3rd!", "Taking the lead!", "You're in 1st!", "Wrong way!"), "Race complete", "New record", "Shortcut!".
- **Character barks:** each playable character has 10+ barks (pickup, hit, boost, overtake, being overtaken, victory, defeat, taunt, ready, start) in a distinct voice and pitch band by class (Light higher, Heavy lower).
- Voice lines have cooldowns (min 2.5 s between barks, no repeats back-to-back) and priority (announcer > player bark > rival bark).

14.7 Ambience and crowd

Each track has an ambience bed (loop) with stereo spots: meadow birds and breeze, harbor gulls and horns, desert wind, snow wind and cable-car creak. Crowds swell at start, finish and for overtakes, with gasps on big hits.

14.8 Dynamic music

Each track and the menu has an original score at the track's BPM (see 04-tracks.md), delivered as **six synchronised stems**: *drums, bass, chords, lead, counter (arp/pad), fx/percussion*. All stems share a loop length of 64 bars worth of material in two sections (A and B).

Game state	Music response
Pre-race	Chords + counter only, quiet
Race start	All stems enter on the downbeat of the GO
Position 1 to 3	full mix, lead + counter prominent
Position 4 to 8	drums+bass+chords, lead softened
Position 9 to 12	add urgent perc/fx and a +2 dB push to drums
Boost / star	filter open, lead +3 dB
Hit / spin-out	low-pass sweep for 0.8 s, then return
Final lap	tempo ramps +6% over 4 s, lead octave lift, fx stem added
Finish	cut to the result jingle (original), then a calm loop

Transitions are beat-quantised (stem gains ramp on the next beat). Stems are exported as lossless 48 kHz / 24-bit masters (high-quality pack) and as AAC for the standard pack.

14.9 Audio performance and QA

- Standard pack: AAC music (about 64 kbps per stem), PCM SFX at 32 kHz, voices at 24 kHz (core under 6 MB on first load, music loaded per track).

- High-quality pack: FLAC 48 kHz/24-bit stems and masters, on demand per track and per character.
 - Test: decode all assets on load; headless AudioContext checks; loudness (EBU R128) check; clipping scan; ducking verification.
-

15. Live-ops, safety and monetisation (summary)

- Seasonal cosmetic passes (free + optional paid cosmetics only), daily and weekly challenges, community ghost races.
 - Child-safe: no chat (preset emotes only), no open voice, private code rooms.
 - Telemetry: opt-in, anonymous, performance and funnel only.
 - Cheating: server-validated results for ranked/daily boards.
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16. Cross-references

- Roster: [02-roster.md](#) · Karts and mods: [03-kart-and-mod-matrix.md](#) · Tracks: [04-tracks.md](#)
- Vertical slice: [05-vertical-slice-plan.md](#) · Production, risks, performance budget: [06-production-risks-perf.md](#)

Roster: 100 Original Characters

Ten themed sets of ten. Every character has a name, a class (Light / Medium / Heavy), a one-line personality, an idle animation, a victory pose, a select-screen portrait and a **distinct driving silhouette** (a head/body shape hook readable at thumbnail size). All characters are original creations.

Class split

Class	Count	Gameplay role
Light	33	Fast acceleration, quick spin-out recovery, weak in bumps
Medium	34	Balanced
Heavy	33	Slow acceleration, high top speed, strong bump

Total: 100 characters, 100 unique names, 12 free starters.

Free starters (12)

Name	Class	Set
Pip Thistledown	Light	Meadow Folk
Bramble Quill	Medium	Meadow Folk
Marigold Hoofsworth	Heavy	Meadow Folk
Fennel Vix	Light	Meadow Folk
Hobb Mossback	Heavy	Meadow Folk
Juniper Wren	Light	Meadow Folk
Clover Dash	Medium	Meadow Folk
Sage Willowmere	Medium	Meadow Folk
Barnaby Bruin	Heavy	Meadow Folk
Captain Dusk Marlowe	Medium	Neon Harbor
Gus Gantry	Heavy	Neon Harbor
Pearl Quayside	Light	Neon Harbor

Unlock methods

- **Coins** (about 60 characters): price = 500 + 100 × (set – 1) + 60 × (position in set – 1).
- **Trophies**: position 5 in each set requires a gold trophy in a named cup (cups rotate).

- **Daily challenge streak:** position 8 in each set unlocks after a streak of 3 + set number days.
- **Achievements:** position 10 in each set (the set "capstone") requires a set-specific achievement:
- Set 1 (Meadow Folk): Win 50 races
- Set 2 (Neon Harbor): Land 500 drift boosts
- Set 3 (Frost Peak): Hit 100 rivals with items
- Set 4 (Dune Runners): Collect 5,000 coins
- Set 5 (Sky Circus): Use the shortcut on every Seedling and Copper track
- Set 6 (Deep Reef): Finish a Grand Prix without being hit
- Set 7 (Clockwork Guild): Win a Versus race with 4 humans
- Set 8 (Ember Forge): Perform 100 slingshots from a draft
- Set 9 (Spirit Grove): Land 25 perfect starts
- Set 10 (Star Cadets): Win a gold Time Trial on 5 different tracks

Voices: each character has 10+ barks in a distinct voice (Light = higher pitch band, Heavy = lower), see the audio section of the GDD.

Set 1: Meadow Folk

Woodland scouts, bakers and tinkers from the Buttercup valleys. Warm earth tones, soft shapes, round silhouettes.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
1	Pip Thistledown ★	Light	Bubbly scout who narrates everything she does out loud	hops in place, straightening her acorn cap	cartwheel while petals burst from her satchel	tall acorn cap with a feather tuft	Free starter
2	Bramble Quill ★	Medium	Grumpy hedgehog mechanic who secretly loves the cheering	polishes his spiky helmet with a rag	takes a deep bow and fans his quills like a peacock	spiky fan-shaped helmet	Free starter
3	Marigold Hoofsworth ★	Heavy	Gentle giant deer who bakes and hugs every rival after the race	chews a pastry, ears twitching	lifts the trophy overhead on her antlers	wide antler crown with ribbons	Free starter
4	Fennel Vix ★	Light	Sly fox courier who always claims she planned it	flicks her tail, checks a pocket watch	spins her tail like a fan and winks	huge swooshing tail	Free starter
5	Hobb Mossback ★	Heavy	Slow-talking tortoise farmer, completely unshakable	tucks head in and out of the shell	shell spins like a top with the trophy on it	domed moss-covered shell backpack	Free starter
6	Juniper Wren ★	Light	Songbird sprinter who hums the whole race	taps the steering wheel in rhythm	flaps up and lands on the kart nose, singing	tiny wings flared at the shoulders	Free starter
7	Clover Dash ★	Medium	Over-caffeinated bunny who speed-reads the map	bounces on her toes	triple back-flip with a fist pump	long ears streaming behind	Free starter
8	Otis Burrowby	Medium	Nearsighted mole engineer with unexpected genius	pushes thick goggles up his nose	pops out of a molehill holding the trophy	goggles and oversized digging claws	Daily challenge streak: 4 days
9	Sage Willowmere ★	Medium	Calm owl tactician who whispers race advice to himself	swivels his head almost all the way round	glides a slow victory loop with wings out	big round facial disc and ear tufts	Free starter
10	Barnaby Bruin ★	Heavy	Honey-loving bear who befriends every rival by lap two	licks honey off a paw	bear-hugs the trophy and a nearby official	round ears and a broad belly	Free starter

Set 2: Neon Harbor

Dockworkers, ferry kids and night-shift legends of the lantern-lit port.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
11	Captain Dusk Marlowe ★	Medium	Weathered tug pilot who calls everyone kid	leans on the wheel and sips tea	tips his cap as an air-horn sounds	peaked captain's cap and pipe	Free starter
12	Roxy Rivet	Light	Welder punk who throws sparks when she gets mad	flips her welding mask up and down	torch-spark fireworks over her head	welding mask and a green mohawk	660 coins
13	Gus Gantry ★	Heavy	Patient crane operator who plans three turns ahead	swings an imaginary crane hook	picks up his own kart with a cheer (hook animation)	broad yellow hard hat and hi-vis vest	Free starter
14	Neona Wavelength	Light	Radio DJ who narrates her own race live	adjusts oversized headphones	drops an imaginary mic	huge neon headphones	780 coins
15	Tobias Tidewell	Medium	Fishmonger with an endless supply of puns	tosses a fish from hand to hand	catches the fish and bows	rubber apron and tall boots	Trophy: gold in the Copper Cup (150cc or higher)
16	Mei Lanternfield	Medium	Quiet lighthouse keeper who is secretly a speedster	scans the horizon with a spyglass	sweeps a beam of light across the crowd	paper lantern on a backpack pole	900 coins
17	Bolt Braddock	Heavy	Retired wrestler turned stevedore who is soft on the inside	flexes one arm then the other	pins the trophy overhead like a champion belt	barrel chest and tiny helmet	960 coins
18	Pearl Quayside ★	Light	Ferry-boat kid with a heart of gold and a fast hand	juggles three coins	coin shower from her oversized coat	tiny frame in a huge coat	Free starter
19	Salvo Reyes	Heavy	Cargo-ship cook who is very proud of his chili	stirs a pot balanced on the kart	ladles trophy chili for the camera	chef hat and ladle	1,080 coins
20	Dmitri Ironwake	Heavy	Icebreaker engineer with a dry sense of humour	folds his arms and nods slowly	shows a rare smile and a thumbs-up	fur-trimmed hat and a thick beard	Achievement: Land 500 drift boosts

Set 3: Frost Peak

Guides, skaters and mountain rescuers of the high white passes.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
21	Nanuk Snowdrift	Heavy	Gentle polar guide who never loses his temper	stretches and yawns	makes a snow angel on the podium	huge hooded shoulders	700 coins
22	Flurry Skye	Light	Showoff ice skater who treats every corner as a routine	spins on one foot	lands a triple axel pose	long trailing scarf	760 coins
23	Grumbald the Yeti	Heavy	Shy yeti who is scared of going fast and wins anyway	hides his face in both hands	roars then waves bashfully	shaggy fur mound with tiny eyes	820 coins
24	Aurora Vance	Medium	Aurora-chasing photographer always framing the shot	frames the scene with her fingers	takes a victory selfie	camera strap and bobble hat	880 coins
25	Tundra Tuck	Medium	Sled-dog musher who talks to his kart like a team	whistles for his dogs	howls at the sky	dog-ear hat and long mittens	Trophy: gold in the Tempest Cup (150cc or higher)
26	Sigrid Hailstone	Heavy	Curling champion who treats racing as strategy	polishes a curling stone	slides the stone down the podium	horned helmet and braids	1,000 coins
27	Pico Sleet	Light	Penguin delivery kid with a very tight schedule	waddles side to side	belly-slide across the finish line	tiny flippers and an orange scarf	1,060 coins
28	Wintra Glimmer	Light	Frost-mage apprentice whose hiccups freeze things	hiccups tiny snowflakes	crowns herself with ice	oversized pointy hat	Daily challenge streak: 6 days

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
29	Kodiak Rime	Medium	Ski-patrol rescuer who always has a plan B	checks a radio on her shoulder	gives two thumbs up	orange vest and goggles	1,180 coins
30	Hoarfrost Hattie	Medium	Knitting granny with a thermos and a lead foot	knits while sitting at a red light	knits a cozy for the trophy	silver bun and a shawl	Achievement: Hit 100 rivals with items

Set 4: Dune Runners

Caravan navigators, scarab-folk and sun-baked traders of the Mirage Mesa.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
31	Zahra Sandglass	Medium	Caravan navigator who has never once been lost	checks a tiny sundial	spins up a swirl of sand	long flowing headscarf	800 coins
32	Rafiq Mirage	Light	Smooth-talking trader who bargains for every position	jingles a pouch of coins	sells the trophy back to the crowd	layered scarves and a feathered hat	860 coins
33	Scarab Sol	Heavy	Beetle-folk bulldozer who pushes everything forward	rolls a small ball with his feet	spins the golden ball on one horn	gleaming shell and a single horn	920 coins
34	Dune Dahlia	Light	Sand-surfing daredevil who hates paved roads	balances on one foot	surfs the podium on a board	goggles and a flapping cape	980 coins
35	Khepri Okoro	Medium	Archaeologist who narrates the history of every track	brushes dust off a relic	holds a golden relic aloft	pith helmet and a satchel	Trophy: gold in the Zenith Cup (150cc or higher)
36	Tamarind Khan	Heavy	Spice merchant sized like a wardrobe, scent of cinnamon	sniffs a pinch of spice and nods	sprinkles spice like confetti	wide sash and heavy shoulders	1,100 coins
37	Sirocco Jade	Light	Wind-sprite who gets bored if she is not in front	drifts in tiny circles	whirlwind exit with a laugh	swirling wispy tail	1,160 coins
38	Nomad Nasir	Medium	Camel-rider turned racer, talks to his camel kart	pats the kart hood	tips his hat to the camel	tall hat with a tassel	Daily challenge streak: 7 days
39	Cactus Carla	Heavy	Prickly saguaro-folk who is surprisingly cuddly	waters herself from a can	bursts into a flower bloom	green arms raised above her head	1,280 coins
40	Ember Anubi	Medium	Jackal-eared night guide who drives better in the dark	ears swivel to a distant sound	howls and strikes a regal pose	tall pointed ears and gold collar	Achievement: Collect 5,000 coins

Set 5: Sky Circus

Acrobats, clowns and balloon pilots from the floating Cloudtop Carnival.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
41	Bunting Barnstorm	Light	Wing-walker pilot who is never quite on the ground	stretches out arms like wings	wing-walks on the kart hood	aviator cap with trailing scarf	900 coins
42	Madame Marzipan	Medium	Ringmistress who announces her own overtakes	twirls a baton	takes a theatrical bow	tall hat with a feather plume	960 coins
43	Tumble Jo	Light	Acrobat who can do a handstand in any situation	does a one-hand handstand	flip into a split	ribbon-streamers on wrists	1,020 coins
44	Big Top Boris	Heavy	Strongman with a gentle soul and a tiny moustache	lifts a dumbbell with a pinky	lifts the kart overhead	curled moustache and striped singlet	1,080 coins
45	Pierrot Fizz	Medium	Mime who communicates only through sound effects	pretends to be trapped in a box	silent applause to the audience	white face and ruff collar	Trophy: gold in the Seedling Cup (150cc or higher)

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
46	Popcorn Pete	Heavy	Concession vendor who throws snacks during overtakes	shakes a popcorn box	throws popcorn everywhere	giant popcorn hat	1,200 coins
47	Lulu Lollipop	Light	Candy-floss artist who sticks to the racing line	twirls candy floss	candy confetti and a spin	big pink swirl hair	1,260 coins
48	Kite Kasumi	Light	Kite-flyer who is fastest on the straights	checks wind direction with a wet finger	launches a kite from the podium	large diamond kite on her back	Daily challenge streak: 8 days
49	Zeppelin Zed	Heavy	Blimp captain who drifts a little too wide	adjusts a brass compass	inflates and floats a foot off the ground	round pilot cap and heavy goggles	1,380 coins
50	Juggles McGraw	Medium	Juggler who treats items like a ball routine	juggles three balls	juggles the trophy	three-pointed jester hat	Achievement: Use the shortcut on every Seedling and Copper track

Set 6: Deep Reef

Divers, mer-folk and glowing creatures from the Coral Causeway.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
51	Coral Calloway	Medium	Marine biologist who narrates the wildlife of every track	taps a clipboard	holds up a glowing jelly	snorkel and fin-shaped helmet	1,000 coins
52	Finn Barracuda	Light	Fast-talking barracuda who loves slipstreaming	swishes his tail fin	jumps and spins mid-air	sleek dorsal fin crest	1,060 coins
53	Octavia Inkwell	Medium	Octopus-lady who handles every control at once	eight arms tapping eight buttons	ink-cloud exit	eight curled tentacle arms	1,120 coins
54	Brinley Bubbles	Light	Pufferfish kid who inflates when startled	puffs up and down	inflates into a round ball	round puffed cheeks	1,180 coins
55	Moby Quill	Heavy	Gentle whale-folk and the largest racer on the grid	blows a spout	breaches the podium with a splash	wide forehead and a small spout	Trophy: gold in the Copper Cup (150cc or higher)
56	Kelp Kelpie	Medium	Sea-spirit who drifts around corners like a current	sways with imaginary tide	rises in a bubble column	green hair that waves like kelp	1,300 coins
57	Urchin Ulla	Heavy	Sea-urchin bruiser whose spines can poke	spins slowly to show off spines	curls into a spiky ball	spiky dome body	1,360 coins
58	Manta Marisol	Light	Manta ray glider who rarely touches the ground	flaps her winglike arms	glides over the podium	wide triangular cape-wings	Daily challenge streak: 9 days
59	Shrimp Sampson	Light	Tiny mantis shrimp with a big punch of attitude	shadowboxes	throws a spinning punch at the camera	boxing-glove claws	1,480 coins
60	Anchor Annabel	Heavy	Retired sailor who steers like a ship	polishes a brass anchor	plants the anchor in the podium	big anchor strapped to her back	Achievement: Finish a Grand Prix without being hit

Set 7: Clockwork Guild

Automata, tinkerers and brass gadgeteers of the Clockwork Quarter.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
61	Cogsworth Whirr	Medium	Prim clockwork butler who is always exactly on time	checks a pocket watch	ticks and bows precisely	top hat with gear-shaped brim	1,100 coins

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
62	Tick-Tock Tilly	Light	Spring-loaded automaton who never stops bouncing	bounces on a spring	launches off the podium in a spiral	coiled spring torso	1,160 coins
63	Gearhart Magnus	Heavy	Steam-powered titan with a soft spot for cats	exhales steam from his pipes	puffs a steam cloud into a heart	chimney stacks on his shoulders	1,220 coins
64	Rivet Rosalind	Medium	Ace mechanic who repairs her kart mid-race	twirls a wrench	holds up a tiny repaired trophy	tool belt and welding goggles	1,280 coins
65	Sprocket Sprout	Light	Plant-powered automaton gardener	waters a pot on his head	flowers bloom from his pot	flower pot on head	Trophy: gold in the Tempest Cup (150cc or higher)
66	Valve Valentina	Medium	Steam-valve engineer with dramatic flair	turns a valve with a hiss	fires a puff of steam fireworks	copper pipes on her backpack	1,400 coins
67	Brass Bartholomew	Heavy	Heavy brass bruiser who talks like a stern grandfather	lowers a monocle	polishes his brass belly	round brass body with a monocle	1,460 coins
68	Pendulum Pia	Light	Clockmaker whose rhythm never wavers	swings gently like a pendulum	swings a watch chain with a wink	oversized pendulum charm	Daily challenge streak: 10 days
69	Dynamo Dex	Medium	Lightning-powered tinkerer with static hair	static crackles in his hair	stands with arc-lightning behind him	spiky hair and twin sparking rods	1,580 coins
70	Lathe Lenore	Heavy	Master woodworker robot who builds a barricade every race	chisels a small figure	presents a carved trophy	wooden-block body with a chisel	Achievement: Win a Versus race with 4 humans

Set 8: Ember Forge

Smiths, salamanders and volcano-dwellers working the Magma Foundry.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
71	Cinder Calhoun	Heavy	Blacksmith who is calm until his hammer is lost	taps his hammer on the kart	hammers an anvil in rhythm	leather apron and thick gloves	1,200 coins
72	Pyra Ashgrove	Light	Fire-sprite who runs hot and fast and cools off with a laugh	flickers like a candle	bursts into harmless flame	flame crest hair	1,260 coins
73	Magma Magnolia	Heavy	Lava-woman with a deep sleepy voice	cracks her knuckles with molten glints	erupts into a glowing pose	orange cracked skin patterns	1,320 coins
74	Slag Wilhelm	Heavy	Furnace-keeper who whistles Irish tunes	shovels imaginary coal	spreads his arms and sings	coal-shovel on his back	1,380 coins
75	Obsidian Orin	Heavy	Silent obsidian golem who communicates by shrugs	shrugs slowly	polished shine pose	glassy black faceted shoulders	Trophy: gold in the Zenith Cup (150cc or higher)
76	Sparkle Salamandra	Light	Salamander racer who sprints on hot floors	scurries on the spot	tail-whip fire loop	spiky tail with ember tip	1,500 coins
77	Vulcana Rex	Heavy	Volcano-guardian queen who adores applause	waves like royalty	cinders rain as a crown	crown-like lava spikes	1,560 coins
78	Anvil Annika	Medium	Apprentice smith who tests every upgrade herself	hammers a small bolt	holds up a forged medal	big hammer slung on shoulder	Daily challenge streak: 11 days
79	Brazier Boone	Light	Torch-bearer who is always the first to arrive	waves a torch	torch relay pose	torch held high	1,680 coins
80	Smelter Sunny	Medium	Cheerful ore-smelter who goes by the nickname Sunny	sorts shiny stones	shiny nugget toss	hard hat with a head-lamp	Achievement: Perform 100 slingshots from a draft

Set 9: Spirit Grove

Friendly ghosts, folklore beings and forest spirits of Moonlit Orchard.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
81	Wisp Willowby	Light	Friendly will-o'-the-wisp who guides lost racers	floats gently	glows brightly with swirling sparkles	glowing orb head	1,300 coins
82	Pumpkin Patch Patsy	Medium	Jack-o-lantern farmer with a glowing grin	tilts her pumpkin head	lantern flare with a laugh	pumpkin head with a stem hat	1,360 coins
83	Spectre Sir Reginald	Medium	Posh ghost butler who haunts the starting grid politely	adjusts his tie	fades out and reappears with the trophy	floating sheet-like cloak	1,420 coins
84	Banshee Bellflower	Light	Songbird spirit whose voice shatters glass but not racers	sings softly	hits a high note that sparkles	long flowing hair	1,480 coins
85	Mothra Moonbeam	Light	Moth sprite attracted to every light on the track	flutters her wings	dusts silver scales	large patterned wings	Trophy: gold in the Seedling Cup (150cc or higher)
86	Hollow Hank	Heavy	Giant scarecrow who is more afraid of crows than karts	slumps on his pole	flaps his arms and crows scatter	straw hat and patched coat	1,600 coins
87	Gourd Granpa	Heavy	Old pumpkin sage who talks in riddles	strokes a pumpkin beard	rolls triumphantly	huge pumpkin with beard	1,660 coins
88	Poltergeist Polly	Medium	Prankster ghost who rearranges the podium	tosses a tiny object	rearranges the podium	ragged ghost tail and bow	Daily challenge streak: 12 days
89	Thistle Thorne	Medium	Forest spirit with vines who grows plants when happy	vines sway	flowers sprout all around	thorny crown of vines	1,780 coins
90	Lantern Lenny	Light	Little lantern spirit who takes the night shift seriously	swings gently	flame flares	lantern for a head	Achievement: Land 25 perfect starts

Set 10: Star Cadets

Rookie pilots, aliens and robots of the Starlight Spaceport academy.

#	Name	Class	Personality	Idle	Victory pose	Silhouette hook	Unlock
91	Nova Starling	Light	Hotshot academy cadet who wants first place at any cost	checks her wrist computer	salutes with a rocket flare	visor helmet with a star decal	1,400 coins
92	Commander Quasar	Heavy	Veteran captain who barks orders at his kart	folds arms and nods	holds a flag with stars	tall command cap and cape	1,460 coins
93	Zorp Blip	Medium	Tiny alien pilot who speaks only in bleeps	bobs antennae	beam-up pose with sparkles	three tiny antennae	1,520 coins
94	Luna Lovelace	Medium	Moon-base coder who debugs her kart mid-corner	types on a floating screen	code sparkles and a bow	visor with scrolling code	1,580 coins
95	Orbit Otto	Heavy	Satellite mechanic who is never quite stationary	spins slowly in place	orbits the podium once	satellite dish on his back	Trophy: gold in the Copper Cup (150cc or higher)
96	Cosmo Chen	Light	Intergalactic courier who has to deliver on time	taps his wrist	spins a delivered parcel	jetpack and small helmet	1,700 coins
97	Pulsar Pippa	Light	Star-kid who glows brighter when she is winning	twinkles quietly	bursts into starlight	star-shaped hair and glowing eyes	1,760 coins
98	Gravity Gloria	Heavy	Gravity-wrangler who changes weight at will	floats half an inch off the ground	floats the trophy overhead	massive shoulder pads and floating rings	Daily challenge streak: 13 days
99	Meteor Mack	Heavy	Rock-skin alien who loves crashing in dramatic style	bumps fists together	crashes through the podium in a cartoon way	cratered round head	1,880 coins
100	Eclipse Elara	Medium	Mysterious pilot of two moons who drives only at night	twirls two small orbs	shows both orbs together	crescent-shaped helmet crest	Achievement: Win a gold Time Trial on 5 different tracks

★ = free starter.

Kart and Modification Matrix

42 kart bodies in 6 families of 7, plus separate wheel sets (18 styles × 5 sizes × 18 colours × 3 finishes), 12 spoilers, 10 exhausts, 8 bumpers, 24 decals and 32 paint colours × 5 finishes with 8 two-tone layouts. Every kart can be modified. Every combination is legal.

Stat model

Stats are 1 to 10: **S** top speed, **A** acceleration, **H** handling, **G** drift grip, **W** weight. **Rule: every body has S + A + H + G = 22** (no body is strictly better). Weight is free (1 to 10) and trades bump strength against acceleration feel.

Final stat = body + character class modifier + wheel + wheel size + spoiler + exhaust + bumper. The sum of mods on each stat is capped at ±1.5. Paint, decals and rim colour never change stats.

Stat	Formula	Range
Top speed (S)	$34 + 1.2 \cdot S \text{ m/s}$	35.2 to 46.0 m/s
Acceleration (A)	$t_{90} = 6.4 - 0.38 \cdot A \text{ s}$	6.0 s to 2.6 s
Handling (H)	yaw cap scaled $(0.72 + 0.03 \cdot H)$	tighter turns
Grip (G)	lat. accel $18 + 1.6 \cdot G \text{ m/s}^2$	19.6 to 34 m/s ²
Weight (W)	mass $0.7 + 0.12 \cdot W$	0.82 to 1.9

Kart bodies

#	Body	Family	S	A	H	G	W	Sum(S+A+H+G)	Top speed m/s (Medium class)	0 to 90% s	Price (coins)	Description
1	Corsa Standard	Cruiser	6	5	6	5	5	22	41.2	4.50	Free	The all-rounder every driver learns on: a red open-wheel with a friendly nose
2	Wayfarer	Cruiser	5	6	5	6	5	22	40.0	4.12	300	Roadster with a small windscreen and a luggage rack
3	Trailblazer	Cruiser	5	5	5	7	6	22	40.0	4.50	450	Raised suspension and a roll hoop, confident on dirt
4	Cobblestone	Cruiser	6	4	6	6	6	22	41.2	4.88	500	Stout tourer with chunky fenders
5	Daybreak	Cruiser	6	6	5	5	4	22	41.2	4.12	550	Sunrise-yellow coupe with a long hood
6	Marlin GT	Cruiser	7	5	5	5	5	22	42.4	4.50	700	Fish-shaped grand tourer with a gill vent
7	Hatchling	Cruiser	4	7	6	5	3	22	38.8	3.74	350	Egg-round little hatchback that leaps off the line
8	Needle	Dart	9	3	5	5	3	22	44.8	5.26	600	Ultra-narrow dart that wants a long straight
9	Javelin	Dart	8	4	5	5	4	22	43.6	4.88	700	Spear-nosed racer with a single tail fin
10	Peregrine	Dart	8	3	6	5	2	22	43.6	5.26	900	Falcon-winged lightweight with swept canopies

#	Body	Family	S	A	H	G	W	Sum(S+A+H+G)	Top speed m/s (Medium class)	0 to 90% s	Price (coins)	Description
11	Comet Tail	Dart	9	4	4	5	4	22	44.8	4.88	1,100	Long-tailed streamliner that leaves a bright wake
12	Silverfin	Dart	7	4	6	5	4	22	42.4	4.88	800	Silver sports car with a shark-fin roof
13	Vandal	Dart	8	5	3	6	5	22	43.6	4.50	950	Rough street-racer with spray-paint trim
14	Rapier	Dart	7	3	7	5	2	22	42.4	5.26	1,000	Needle-nosed fencer that carves tidy lines
15	Slidewinder	Drifter	4	5	6	7	4	22	38.8	4.50	350	Low serpent body with a spiral exhaust
16	Pirouette	Drifter	3	5	8	6	2	22	37.6	4.50	900	Ballerina-shaped lightweight, precise on tight turns
17	Skiff	Drifter	4	4	7	7	3	22	38.8	4.88	650	Boat-hull kart that glides sideways
18	Whirligig	Drifter	3	6	7	6	3	22	37.6	4.12	800	Spinning-top body with a rotating roof fan
19	Tango	Drifter	5	4	6	7	4	22	40.0	4.88	700	Two-tone dancer with a flared tail
20	Chicane	Drifter	4	5	5	8	5	22	38.8	4.50	1,000	Rally-bred car with huge side skirts
21	Cornerstone	Drifter	5	3	7	7	4	22	40.0	5.26	1,200	Wedge-shaped tight-corner specialist
22	Ironclad	Bruiser	7	2	4	9	10	22	42.4	5.64	1,000	Armour-plated brick on tracks, nothing moves it
23	Bulwark	Bruiser	6	3	5	8	9	22	41.2	5.26	1,100	Shield-nosed tank with side bumpers
24	Mammoth	Bruiser	8	2	3	9	10	22	43.6	5.64	1,500	Tusked front bumper, enormous rear tyres
25	Rhino Cab	Bruiser	7	3	4	8	9	22	42.4	5.26	1,300	Horned taxi cab that clears the road
26	Dozer	Bruiser	5	4	4	9	8	22	40.0	4.88	900	Yellow bulldozer blade, slow to start but unstoppable
27	Brickhouse	Bruiser	6	4	4	8	9	22	41.2	4.88	1,000	Boxy brick kart with a chimney exhaust
28	Gantry Hauler	Bruiser	7	2	5	8	10	22	42.4	5.64	1,400	Cargo hauler with a flatbed and crane arm
29	Pogo	Rocket	3	9	5	5	2	22	37.6	2.98	Free	Spring-loaded pogo kart that explodes off the line
30	Sparkplug	Rocket	4	8	5	5	3	22	38.8	3.36	400	Compact hot-rod with exposed engine
31	Kickstart	Rocket	4	8	4	6	4	22	38.8	3.36	600	Motorcycle-engine kart with visible chain drive
32	Bottle Rocket	Rocket	5	9	3	5	3	22	40.0	2.98	1,000	Bottle-shaped body with a flame nozzle
33	Ignition	Rocket	5	7	5	5	4	22	40.0	3.74	700	Muscle car with twin tailpipes
34	Hotshot	Rocket	4	7	6	5	3	22	38.8	3.74	800	Flame-decal short-wheelbase racer
35	Lift-off	Rocket	3	8	6	5	2	22	37.6	3.36	1,100	Capsule body with a tiny rocket booster
36	Tubby Tub	Oddball	5	5	3	9	7	22	40.0	4.50	800	Bathtub on four wheels with a rubber duck
37	Pumpkin Coach	Oddball	6	4	6	6	6	22	41.2	4.88	1,000	Carved pumpkin carriage with lantern lights
38	Teacup Twister	Oddball	3	6	9	4	2	22	37.6	4.12	900	Giant spinning teacup that turns on a dime
39	Mail Run	Oddball	6	6	4	6	5	22	41.2	4.12	600	Rural mailbox on wheels with a flag that pops up on boosts
40	Sneaker Sled	Oddball	4	7	7	4	3	22	38.8	3.74	750	Oversized sneaker with a laced-up cockpit
41	Cheese Wedge	Oddball	8	5	4	5	4	22	43.6	4.50	1,200	Triangular cheese wedge with holes and a mouse mascot
42	Hover Pancake	Oddball	5	3	8	6	1	22	40.0	5.26	1,500	Floating stack of pancakes on small hover pads

Family traits

Family	Identity	Typical strength	Typical weakness
Cruiser	Friendly all-rounders	balanced	no standout
Dart	Spear-nosed speedsters	top speed	slow acceleration
Drifter	Low wedges and boats	handling + grip	low top speed
Bruiser	Armoured bricks	weight, top speed	slow acceleration, poor handling
Rocket	Bottles, flames, springs	acceleration	handling
Oddball	Props and fun shapes	unique trade-offs	uneven

Free starter bodies: Corsa Standard, Pogo.

Wheels

18 rim styles. Deltas below are for the 14" size; apply the size table on top.

Style	Tire compound	S	A	H	G	W	Off-road	Description	Price
Six-Spoke Standard	Street	+0	+0	+0	+0	+0	+0	Classic six-spoke alloy	Free
Turbine Fan	Street	+0.3	-0.2	+0	+0	+0	+0	Fan-blade rim, tuned for top speed	250
Mesh Classic	Street	+0	+0.2	+0	-0.1	-0.2	+0	Lightweight mesh rim	250
Dish Deep	Racing	+0.2	+0	+0.1	-0.1	+0.2	+0	Deep-dish chrome look	350
Starburst	Street	+0	+0.1	+0.2	+0	+0	+0	Ten-spoke star rim	300
Slick Racing	Slick	+0.2	+0	+0	+0.5	+0	-0.5	Slick compound, loves tarmac, hates dirt	500
Trail Grip	Trail	-0.2	+0	+0	+0.1	+0.2	+0.6	Knobbly trail tyres for shortcuts	450
Mudder	Mud	-0.3	+0	+0	+0.2	+0.4	+1	Deep-lug mud tyres, best off-road	600
Balloon Soft	Balloon	-0.3	+0.1	+0.2	+0	+0.2	+0.3	Soft balloon tyres that absorb bumps	400
Featherweight	Carbon	+0.1	+0.4	+0	-0.2	-0.5	-0.2	Carbon rims, ultra light	700
Anvil Steel	Street	+0	-0.3	+0	+0.2	+0.5	+0.1	Heavy steel rims, planted	400
Rally Grip	Trail	+0	+0.1	+0.1	+0.3	+0	+0.4	Rally compound all-rounder	550
Ice Studs	Studded	-0.1	+0	+0	+0.3	+0.1	+0.2	Studded for slippery ice sections	500
Hover Pad	Hover	-0.2	+0.2	+0.3	-0.2	-0.6	+0.5	Short hover pads, light but loose	900
Flywheel	Racing	+0.3	-0.1	+0	+0	+0.3	+0	Flywheel-balanced racing rim	650
Spinner Disc	Racing	+0.1	+0.1	+0.1	+0	+0	+0	Disc cover with a spinner	500
Candy Cane	Street	+0	+0	+0.2	+0.1	+0	+0	Striped twisted spoke	350
Gearwheel	Street	+0	+0.2	+0	+0.1	+0.1	+0	Cog-shaped rim, clanks on corners	450

Wheel sizes (all styles)

Size	S	A	H	G	W
12"	-0.4	+0.4	+0.2	+0.1	-0.2
13"	-0.2	+0.2	+0.1	+0.05	-0.1
14"	+0	+0	+0	+0	+0
15"	+0.2	-0.2	-0.1	-0.05	+0.1
16"	+0.4	-0.4	-0.2	-0.1	+0.2

Small wheels accelerate faster and turn tighter; large wheels have a higher top speed. Wheels are swappable on any kart. "Off-road" modifies the off-road top-speed multiplier by that fraction of the penalty recovered (+1.0 recovers 25 points of the 45% slow-down on grass).

Rim colours and finishes (cosmetic)

Colours: Chrome, Gunmetal, Gold, Bronze, Signal Red, Cherry, Sunset Orange, Lemon, Lime, Mint, Teal, Sky Blue, Cobalt, Violet, Hot Pink, Pearl White, Jet Black, Rainbow Anodized.

Finishes: Polished, Satin, Anodized.

Spoilers

Name	S	A	H	G	W	Description	Price
None	+0	+0	+0	+0	+0	No wing	Free
Low Lip	+0.1	+0	+0	+0.1	+0	Small ducktail lip	100
Duck Tail	+0.1	+0.1	+0	+0.1	+0.1	Classic ducktail	150
GT Wing	+0.2	-0.1	+0	+0.3	+0.1	Large wing, more grip at speed	300
Dual Plane	+0.3	-0.2	+0	+0.3	+0.2	Double-element wing for fast tracks	450
Swan Neck	+0.2	+0	-0.1	+0.4	+0.1	Swan-neck mount, strong downforce	500
Barn Door	-0.1	-0.3	+0.1	+0.6	+0.3	Huge barn-door wing, great grip but slow	450
Shark Fin	+0.1	+0.1	+0.3	+0	+0	Vertical fin for crisp handling	350
Roof Scoop	+0	+0.3	+0	+0	+0.1	Intake scoop, helps accel	350
Twin Tail	+0.2	+0.1	+0.1	+0	+0	Twin-tail rudders	500
Pop-up Flap	+0.3	-0.3	+0	+0.1	+0	Air-brake flap used on long straights	400
Feather Wing	+0.1	+0.2	+0	-0.1	-0.3	Carbon feather-light wing	600

Exhausts

Name	S	A	H	G	W	Description	Price
Stock Pipe	+0	+0	+0	+0	+0	Standard single tailpipe	Free
Twin Chrome	+0.1	+0.1	+0	+0	+0.1	Twin chrome pipes	150
Side Pipes	+0	+0.2	+0	+0	+0	Side-exit pipes	200
Megaphone	+0.2	+0.1	+0	+0	+0.1	Loud megaphone exhaust	250
Upswept	+0.1	+0	+0.1	+0	+0	Upswept pipe, tiny gain in clearance	250
Flame Thrower	+0	+0.3	+0	-0.1	+0.1	Shoots flames on boost	400
Quad Stack	+0.3	+0.1	+0	-0.1	+0.2	Four-barrel stack	500
Turbo Whistle	+0.1	+0.3	+0	+0	+0	Whistling turbo exhaust	550
Bubbler	+0	+0.1	+0.1	+0.1	+0	Blows bubbles on boost	400
Rocket Nozzle	+0.4	+0.3	-0.1	-0.1	+0.2	Single rocket nozzle, high power but demanding	700

Bumpers

Name	S	A	H	G	W	Description	Price
Stock Bumper	+0	+0	+0	+0	+0	Standard bumper	Free
Rubber Pusher	+0	+0	+0	+0.1	+0.2	Soft rubber bumper that shrugs off bumps	200
Splitter	+0.1	+0	+0.1	+0	-0.1	Aero splitter	300
Cow Catcher	-0.1	-0.1	+0	+0	+0.4	Heavy cow-catcher grille	350
Spike Guard	+0	+0	+0	+0.1	+0.3	Decorative spikes, increases push-through	400
Tiny Bumper	+0.1	+0.1	+0.1	-0.1	-0.3	Minimal bumper for lightness	450
Rubber Duck Horn	+0	+0.1	+0	+0	+0.1	Squeaky horn bumper (cosmetic horn sound)	250
Twin Prongs	+0.1	+0	+0.2	+0	+0.1	Forked prongs for light contact	500

Paint, decals and finishes (cosmetic only, no stats)

Paint colours (32): Cherry Red, Sunset Orange, Marigold, Lemon Zest, Lime Pop, Meadow Green, Mint Julep, Teal Wave, Sky Blue, Cobalt, Indigo Night, Violet Haze, Orchid, Hot Pink, Bubblegum, Coral, Chocolate, Sandstone, Pearl White, Ice Silver, Gunmetal, Jet Black, Midnight Blue, Forest, Rust, Peach, Aqua, Lavender, Burgundy, Olive, Turquoise, Gold Leaf.

Finishes (5): Gloss, Matte, Metallic, Pearl, Candy (Gloss free; others 300 coins each).

Two-tone layouts (8): Hood Stripe, Split Down, Roof Cap, Fade Front-Back, Racing Number Panel, Bib, Lower Skirt, Diagonal.

Decals (24): Racing Stripes, Twin Stripes, Checker Flag, Lightning Bolt, Flame Licks, Polka Dots, Star Field, Camo Splash, Zigzag, Wave Crest, Honeycomb, Number 7, Number 42, Number 99, Sun Burst, Skull & Wrenches, Paw Prints, Leaf Pattern, Snowflakes, Circuit Lines, Candy Swirl, Tiger Stripes, Argyle, Galaxy Swirl.

Prices: paint colour 50 coins, finish 300, two-tone 200, decal 80.

Which slot affects which stat

Slot	S	A	H	G	W	Notes
Body	•	•	•	•	•	Sets the base; sum of S+A+H+G fixed at 22
Character class	•	•	•	•	•	Light / Medium / Heavy modifiers
Wheel style	•	•	•	•	•	Plus tire compound off-road behaviour
Wheel size	•	•	•	•	•	12" to 16"
Spoiler	•	•	•	•	•	
Exhaust	•	•	•	•	•	

Slot	S	A	H	G	W	Notes
Bumper	•	•	•	•	•	Best for weight
Paint / decal / rim colour						Cosmetic only

Mod extremes (to check the cap)

Stat	Best wheel	Best spoiler	Best exhaust	Best bumper	Total mod
Top speed	Turbine Fan	Dual Plane	Rocket Nozzle	Splitter	+1.50 (capped at ±1.5)
Acceleration	Featherweight	Roof Scoop	Flame Thrower	Tiny Bumper	+1.50 (capped at ±1.5)
Handling	Hover Pad	Shark Fin	Upswept	Twin Prongs	+1.10 (capped at ±1.5)
Grip	Slick Racing	Barn Door	Bubbler	Rubber Pusher	+1.40 (capped at ±1.5)
Weight	Anvil Steel	Barn Door	Quad Stack	Cow Catcher	+1.50 (capped at ±1.5)

Tracks: 20 Original Tracks

Four cups of four tracks plus four bonus tracks. Every track has: a distinct theme, a clear racing line, **at least one risky shortcut**, a hazard, a landmark you can recognise at a glance, item rows that do not spawn unfairly on the first corner (the first row is never within 120 m of the grid and never inside a corner), and a **mirror and reverse** variant.

Lap times are for a Medium-class kart on 150cc with an average-skill driver (length ÷ average speed). Tracks marked as *built* have been measured from the playable web slice's actual track spline.

Overview

#	Track	Cup	Theme	Length	Lap time	3 laps	Music
1	Buttercup Meadows	Seedling Cup	Sunny meadow valley with windmills, sunflower fields and a covered bridge	1,650 m (as built)	0:52	2:35	122 BPM C major
2	Lantern Harbor	Seedling Cup	Night port with paper lanterns, cranes and moored ships	1,980 m (as built)	1:00	3:00	126 BPM A minor
3	Mirage Mesa	Seedling Cup	Desert canyon with sandstone arches and a dry riverbed	2,232 m (as built)	1:04	3:11	118 BPM D dorian
4	Frostbite Pass	Seedling Cup	Snowy mountain pass with an ice cave and a frozen waterfall	2,066 m (as built)	1:01	3:02	110 BPM E minor
5	Clockwork Quarter	Copper Cup	Steampunk city with giant gears and brass pipes	1,320 m	0:43	2:08	132 BPM G minor
6	Coral Causeway	Copper Cup	Underwater glass tunnels over a coral reef	2,520 m	1:14	3:42	116 BPM F lydian
7	Jade Terrace	Copper Cup	Terraced mountain gardens with stone lanterns and a pagoda	2,780 m	1:19	3:58	104 BPM D pentatonic
8	Moonlit Orchard	Copper Cup	Cute spooky orchard with friendly ghosts and lanterns	1,900 m	0:58	2:53	128 BPM B minor
9	Magma Foundry	Tempest Cup	Volcano forge with lava rivers and anvils	2,150 m	1:03	3:10	138 BPM C# phrygian
10	Cloudtop Carnival	Tempest Cup	Sky fairground on clouds with ferris wheel and roller-coaster	2,360 m	1:09	3:28	134 BPM F major
11	Stormwatch Cliffs	Tempest Cup	Coastal cliffs with a lighthouse and an incoming storm	3,100 m	1:24	4:11	112 BPM A minor
12	Glowcap Caverns	Tempest Cup	Crystal caves with bioluminescent mushrooms	1,780 m	0:56	2:47	124 BPM E dorian
13	Starlight Spaceport	Zenith Cup	Launch complex with gantries, fuel tanks and rockets	2,640 m	1:13	3:40	142 BPM F# minor
14	Neon Metro Loop	Zenith Cup	Night city with holograms, a tram and skyscrapers	2,900 m	1:21	4:02	140 BPM G minor
15	Overgrowth Ruins	Zenith Cup	Jungle temple with vines, waterfalls and a stone face	2,430 m	1:09	3:28	120 BPM C phrygian
16	Zenith Ring	Zenith Cup	Anti-gravity ring in the sky around a ringed planet	3,400 m	1:30	4:28	150 BPM A minor
17	Rustbelt Rally	Bonus	Industrial scrapyards with crushed-car mountains	1,090 m	0:34	1:42	130 BPM B minor
18	Sugarplum Speedway	Bonus	Candy land with lollipop trees and chocolate rivers	1,560 m	0:49	2:26	136 BPM D major
19	Polar Lights Circuit	Bonus	Ice rink under the aurora with a snowman crowd	1,250 m	0:40	2:01	114 BPM F# dorian

#	Track	Cup	Theme	Length	Lap time	3 laps	Music
20	Thunderhead Savanna	Bonus	Safari plains with herds, acacia trees and a dust storm	3,300 m	1:29	4:28	108 BPM G mixolydian

Cups

- **Seedling Cup:** Buttercup Meadows, Lantern Harbor, Mirage Mesa, Frostbite Pass
- **Copper Cup:** Clockwork Quarter, Coral Causeway, Jade Terrace, Moonlit Orchard
- **Tempest Cup:** Magma Foundry, Cloudtop Carnival, Stormwatch Cliffs, Glowcap Caverns
- **Zenith Cup:** Starlight Spaceport, Neon Metro Loop, Overgrowth Ruins, Zenith Ring
- **Bonus:** Rustbelt Rally, Sugarplum Speedway, Polar Lights Circuit, Thunderhead Savanna

Shortest track: Rustbelt Rally (1,090 m). Longest: Zenith Ring (3,400 m).

Design rules for every track

1. **Racing line:** the line is readable from the road colour/texture and bright edge markings; no blind apexes in the first three seconds after the start.
2. **Shortcut:** every track has a risky shortcut with a visible entry (arrow + gate or different ground). It saves 1.5 to 3.5 s with a boost and costs time without one.
3. **Hazard:** telegraphed 1 to 2 s before it can hurt; every hazard has a visual language (pulse decal, sound cue).
4. **Item rows:** the first row is at least 120 m after the grid and not in a corner; rows are placed after a corner exit, before shortcuts, and never inside a hazard. Rows are 4 to 6 orbs wide.
5. **Variants:** *Mirror* flips left/right (and the signage). *Reverse* runs the track backwards; start line, rows, hazards and the shortcut entry are re-authored.
6. **Anti-gravity / banked sections** appear only on Zenith Ring and Neon Metro Loop and stay readable with strong glow-line borders.

1. Buttercup Meadows

- **Cup:** Seedling Cup
- **Theme:** Sunny meadow valley with windmills, sunflower fields and a covered bridge
- **Length:** 1,650 m (measured from the built web slice) · **Lap time:** 0:52 · **3 laps:** 2:35
- **Signature shortcut:** Hay-bale gap through the sunflower field behind the windmill; rough ground, a boost pad at the entry saves about 2.5 s
- **Hazard:** Sheep herd crosses the hairpin; a bell rings 2 s before they step out

- **Landmark:** Giant windmill whose blades sweep over the finish straight
- **Item-box notes:** First item row sits 260 m after the grid and spans the full road width on the long first straight; second row is after the covered bridge
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse swaps the hay-bale gap entry to the left-hand side and moves the sheep crossing to the bridge exit
- **Music:** 122 BPM, C major
- **Ambience:** Birdsong, wind, distant cowbells, light crowd on the grandstand

2. Lantern Harbor

- **Cup:** Seedling Cup
- **Theme:** Night port with paper lanterns, cranes and moored ships
- **Length:** 1,980 m (measured from the built web slice) · **Lap time:** 1:00 · **3 laps:** 3:00
- **Signature shortcut:** Slip through the container row; container doors open and close on a 6 s cycle, a closed door costs a wall bump
- **Hazard:** Crane loads swing over the dockside straight, shown by a ground shadow 1.2 s early
- **Landmark:** Lighthouse tower with a rotating beam that lights the track
- **Item-box notes:** Item rows sit on the wide pier after turn 2 and before the container row so a shortcut hopeful can arrive with an item
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse flips the container cycle phase so the doors are closed when you arrive on lap 1
- **Music:** 126 BPM, A minor
- **Ambience:** Gulls, ship horns, water lapping, crowd cheer from pier

3. Mirage Mesa

- **Cup:** Seedling Cup
- **Theme:** Desert canyon with sandstone arches and a dry riverbed
- **Length:** 2,232 m (measured from the built web slice) · **Lap time:** 1:04 · **3 laps:** 3:11
- **Signature shortcut:** Jump the dry riverbed through the twin arch over a sand ramp; needs a mini-turbo to hold speed on the sand
- **Hazard:** Boulders roll from the canyon wall in the narrows, each telegraphed by a dust puff
- **Landmark:** Twin Arch rock formation
- **Item-box notes:** Item rows after the first arch and at the end of the canyon; none within 120 m of the narrows entry
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse moves the boulder gap to the other canyon wall and replaces the sand ramp with a cliff-edge ramp
- **Music:** 118 BPM, D dorian

- **Ambience:** Desert wind, distant hawk cries, rolling dust, low crowd

4. Frostbite Pass

- **Cup:** Seedling Cup
- **Theme:** Snowy mountain pass with an ice cave and a frozen waterfall
- **Length:** 2,066 m (measured from the built web slice) · **Lap time:** 1:01 · **3 laps:** 3:02
- **Signature shortcut:** Ice-cave tunnel through the mountain; slippery surface unless you hold a drift line
- **Hazard:** Icicles drop in the cave with a shadow telegraph; ice patches on the outside bend
- **Landmark:** Frozen waterfall next to a cable car line
- **Item-box notes:** Item rows before the cave and on the downhill straight; no rows inside the cave
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse sends you through the cave uphill with ice patches placed on the new entry
- **Music:** 110 BPM, E minor
- **Ambience:** Howling wind, creaking cable car, snow crunch

5. Clockwork Quarter

- **Cup:** Copper Cup
- **Theme:** Steampunk city with giant gears and brass pipes
- **Length:** 1,320 m · **Lap time:** 0:43 · **3 laps:** 2:08
- **Signature shortcut:** Slip through the gear tower; timed gate opens every 8 s
- **Hazard:** Rotating cog-bars and steam vents at the central square
- **Landmark:** Giant clock tower that chimes on the hour (music syncs)
- **Item-box notes:** Compact track: two item rows only, after the second corner and before the square
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse swaps the gear tower gate to the exit and adds a steam vent at the new entry
- **Music:** 132 BPM, G minor
- **Ambience:** Ticking, steam hiss, distant factory whistles

6. Coral Causeway

- **Cup:** Copper Cup
- **Theme:** Underwater glass tunnels over a coral reef
- **Length:** 2,520 m · **Lap time:** 1:14 · **3 laps:** 3:42

- **Signature shortcut:** Kelp-forest cut-through with current streams that boost your speed
- **Hazard:** Pufferfish mines drift across the road, they inflate when you approach
- **Landmark:** A giant whale gliding overhead
- **Item-box notes:** Rows sit before each glass tunnel exit; one row inside the kelp shortcut for risk-reward
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse flips the current direction in the kelp shortcut so it becomes a slow gap
- **Music:** 116 BPM, F Lydian
- **Ambience:** Muffled water, bubbles, distant whale song

7. Jade Terrace

- **Cup:** Copper Cup
- **Theme:** Terraced mountain gardens with stone lanterns and a pagoda
- **Length:** 2,780 m · **Lap time:** 1:19 · **3 laps:** 3:58
- **Signature shortcut:** Bamboo bridge between terraces; narrow, no walls
- **Hazard:** Swinging log gate at the temple steps
- **Landmark:** Five-storey pagoda
- **Item-box notes:** Item rows on terraces 1 and 3; none on the bridge
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse shifts the log gate timing and replaces the bridge entrance with a ramp
- **Music:** 104 BPM, D pentatonic
- **Ambience:** Wind chimes, bamboo creak, temple bell

8. Moonlit Orchard

- **Cup:** Copper Cup
- **Theme:** Cute spooky orchard with friendly ghosts and lanterns
- **Length:** 1,900 m · **Lap time:** 0:58 · **3 laps:** 2:53
- **Signature shortcut:** Run through the hollow giant tree; slow bark floor, but it saves a hairpin
- **Hazard:** Mischievous scarecrows lean into the road after a hidden cue
- **Landmark:** Giant gnarled tree glowing with lanterns
- **Item-box notes:** Item rows after the orchard hairpin and before the hollow tree
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse puts the scarecrows on the opposite verge
- **Music:** 128 BPM, B minor
- **Ambience:** Crickets, owls, creaking branches, ghostly giggles

9. Magma Foundry

- **Cup:** Tempest Cup
- **Theme:** Volcano forge with lava rivers and anvils
- **Length:** 2,150 m · **Lap time:** 1:03 · **3 laps:** 3:10
- **Signature shortcut:** Cross the cooled lava crust; it is slow rough ground but cuts a long bend
- **Hazard:** Lava geysers burst with a glowing telegraph; forge hammers swing over the road
- **Landmark:** Colossal anvil statue
- **Item-box notes:** Rows before the forge hall and after the lava bridge
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse moves the geysers to the exit side
- **Music:** 138 BPM, C# phrygian
- **Ambience:** Rumbling, forge clangs, bubbling lava

10. Cloudbottom Carnival

- **Cup:** Tempest Cup
- **Theme:** Sky fairground on clouds with ferris wheel and roller-coaster
- **Length:** 2,360 m · **Lap time:** 1:09 · **3 laps:** 3:28
- **Signature shortcut:** Ride the roller-coaster rail section, boost pad at the entrance
- **Hazard:** Bumper-car clusters shuffle across the road
- **Landmark:** Giant ferris wheel with spinning gondolas
- **Item-box notes:** Rows after the carousel and before the rail shortcut
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse flips the rail direction so it climbs instead of drops
- **Music:** 134 BPM, F major
- **Ambience:** Fair music loops, laughing crowd, clinking rides

11. Stormwatch Cliffs

- **Cup:** Tempest Cup
- **Theme:** Coastal cliffs with a lighthouse and an incoming storm
- **Length:** 3,100 m · **Lap time:** 1:24 · **3 laps:** 4:11
- **Signature shortcut:** Dive into the sea cave beneath the arch; waves surge on a 10 s cycle
- **Hazard:** Lightning strikes on exposed bends, telegraphed by a ring of light
- **Landmark:** Tall lighthouse sending beams into the storm
- **Item-box notes:** Rows at the cliff top and after the coastal bridge

- **Mirrored / reverse variant:** Mirror flips the layout; Reverse changes the wave cycle and puts the lightning ring on the opposite bend
- **Music:** 112 BPM, A minor
- **Ambience:** Crashing waves, thunder, gull cries

12. Glowcap Caverns

- **Cup:** Tempest Cup
- **Theme:** Crystal caves with bioluminescent mushrooms
- **Length:** 1,780 m · **Lap time:** 0:56 · **3 laps:** 2:47
- **Signature shortcut:** Cross the crystal bridge over the chasm; narrow, no walls
- **Hazard:** Stalactites drop with a shadow warning
- **Landmark:** Crystal Cathedral, a huge glowing cavern
- **Item-box notes:** Rows before the cathedral and after the chasm
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse puts the stalactite drops on the exit stretch
- **Music:** 124 BPM, E dorian
- **Ambience:** Dripping, echoing hum, crystal chimes

13. Starlight Spaceport

- **Cup:** Zenith Cup
- **Theme:** Launch complex with gantries, fuel tanks and rockets
- **Length:** 2,640 m · **Lap time:** 1:13 · **3 laps:** 3:40
- **Signature shortcut:** Cut through the launch gantry; gate opens briefly
- **Hazard:** Rocket exhaust blasts across the road on a timer
- **Landmark:** A rocket that launches during lap 3
- **Item-box notes:** Rows after the tank farm and before the gantry
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse swaps the exhaust timing and moves the launch cutscene to lap 2
- **Music:** 142 BPM, F# minor
- **Ambience:** Countdown PA announcements, humming engines, distant crowd

14. Neon Metro Loop

- **Cup:** Zenith Cup

- **Theme:** Night city with holograms, a tram and skyscrapers
- **Length:** 2,900 m · **Lap time:** 1:21 · **3 laps:** 4:02
- **Signature shortcut:** Run through the rail yard under the tram
- **Hazard:** Trams cross the road, indicated by flashing lights and a bell
- **Landmark:** Skyscraper with a holographic dragon
- **Item-box notes:** Rows on the boulevard and after the rail yard
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse makes the tram cross on the other side
- **Music:** 140 BPM, G minor
- **Ambience:** City hum, tram bells, distant sirens, holographic chime

15. Overgrowth Ruins

- **Cup:** Zenith Cup
- **Theme:** Jungle temple with vines, waterfalls and a stone face
- **Length:** 2,430 m · **Lap time:** 1:09 · **3 laps:** 3:28
- **Signature shortcut:** Dash through the collapsed temple hall; rubble slows you
- **Hazard:** Rolling stone drum that rolls down the stairs on a timer
- **Landmark:** Giant stone face carved in the cliff
- **Item-box notes:** Rows before the temple, none inside the hall
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse sends the rolling stone up the stair instead
- **Music:** 120 BPM, C phrygian
- **Ambience:** Jungle birds, waterfall, drums

16. Zenith Ring

- **Cup:** Zenith Cup
- **Theme:** Anti-gravity ring in the sky around a ringed planet
- **Length:** 3,400 m · **Lap time:** 1:30 · **3 laps:** 4:28
- **Signature shortcut:** Take the gravity-flip loop in the inner ring; requires boost to climb
- **Hazard:** Asteroid pulses along the outer ring
- **Landmark:** A ringed planet
- **Item-box notes:** Rows on each ring segment, none on the climb
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse flips the gravity direction so the climb is a drop
- **Music:** 150 BPM, A minor
- **Ambience:** Deep space hum, energy swell, distant announcer

17. Rustbelt Rally

- **Cup:** Bonus
- **Theme:** Industrial scrapyard with crushed-car mountains
- **Length:** 1,090 m · **Lap time:** 0:34 · **3 laps:** 1:42
- **Signature shortcut:** Climb the scrap ramp over the car pile
- **Hazard:** Magnet crane lifts cars across the road
- **Landmark:** Mountain of crushed cars
- **Item-box notes:** Item rows after the crusher
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse swaps the ramp direction
- **Music:** 130 BPM, B minor
- **Ambience:** Clanking, grinding, distant welding

18. Sugarplum Speedway

- **Cup:** Bonus
- **Theme:** Candy land with lollipop trees and chocolate rivers
- **Length:** 1,560 m · **Lap time:** 0:49 · **3 laps:** 2:26
- **Signature shortcut:** Slide down the chocolate slope between candy canes
- **Hazard:** Gumdrops pop out of the sugar
- **Landmark:** Giant lollipop tree
- **Item-box notes:** Item rows on the sugar plaza
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse flips the chocolate slope into a climb
- **Music:** 136 BPM, D major
- **Ambience:** Cheerful chimes, popping candy, crowd

19. Polar Lights Circuit

- **Cup:** Bonus
- **Theme:** Ice rink under the aurora with a snowman crowd
- **Length:** 1,250 m · **Lap time:** 0:40 · **3 laps:** 2:01
- **Signature shortcut:** Cut across the frozen pond infield
- **Hazard:** Ice patches in the corners
- **Landmark:** Aurora ribbons over a giant snowman
- **Item-box notes:** Item rows on the straight
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse moves the frozen pond entry
- **Music:** 114 BPM, F# dorian

- **Ambience:** Soft wind, ice creaks, distant skates

20. Thunderhead Savanna

- **Cup:** Bonus
- **Theme:** Safari plains with herds, acacia trees and a dust storm
- **Length:** 3,300 m · **Lap time:** 1:29 · **3 laps:** 4:28
- **Signature shortcut:** Cross the dry waterhole
- **Hazard:** Herd of wildebeest crosses the road, signalled by dust
- **Landmark:** Giant baobab tree
- **Item-box notes:** Rows on the plains stretch and after the waterhole
- **Mirrored / reverse variant:** Mirror flips the layout; Reverse moves the herd crossing and changes the waterhole entry
- **Music:** 108 BPM, G mixolydian
- **Ambience:** Wind in grass, distant animals, drums

Vertical Slice Plan

Goal: a slice that already feels like the \$10M game, not a prototype. The slice is the quality yardstick: if something is not at this bar in the slice, it is not ready to scale to 100 characters and 20 tracks.

1. Scope

Area	Slice content
Characters	8 playable: Pip Thistledown (L), Fennel Vix (L), Juniper Wren (L), Bramble Quill (M), Clover Dash (M), Captain Dusk Marlowe (M), Marigold Hoofsworth (H), Gus Gantry (H). Four more free starters appear as CPU rivals (Pearl Quayside, Sage Willowmere, Hobb Mossback, Barnaby Bruin) to fill the 12-kart grid.
Karts	6 bodies: Corsa Standard (Cruiser), Needle (Dart), Slidewinder (Drifter), Ironclad (Bruiser), Pogo (Rocket), Pumpkin Coach (Oddball)
Garage	Full: 18 wheel styles × 5 sizes × colours, 12 spoilers, 10 exhausts, 8 bumpers, 32 paint colours × 5 finishes, 24 decals, live stat bars, 3D preview, save/load
Tracks	4: Buttercup Meadows (1650 m), Lantern Harbor (1980 m), Mirage Mesa (2232 m), Frostbite Pass (2066 m) (a Seedling Cup subset: meadow, harbor night, desert, snow), each with a shortcut and a hazard, 3 laps default
Multiplayer	4-player code lobbies (6-character code), host settings (track, laps, items, CPU fill), connection status
Items	All 8 items with the exact behaviours in the GDD
Drift boost	Three tiers + start boost
Audio	Full dynamic score per track and menu, layered engines per class, announcer, barks, ambience, crowd, mixer
Platforms	iOS Safari, Android Chrome (web), desktop browsers; gamepad

2. Quality-bar checklist (all must be true to call the slice "done")

Feel - [] Steering feels responsive at 60 fps; touch slider is accurate to 1° of heading. - [] Drift entry is satisfying: hop, slide, sparks, tier changes are obvious without looking at a bar. - [] Boost feels fast: FOV kick, trail, speed lines, whoosh, camera shake. - [] Every item is understandable in 2 seconds, with a clear counter. - [] CPUs drive plausibly, take shortcuts at higher skill, and never feel scripted.

Visual - [] Every track is recognisable from a single screenshot (landmark visible). - [] Rim light separates karts from the background at any time of day. - [] No placeholder boxes: every object belongs to a theme. - [] Characters readable at thumbnail size.

Audio (top priority) - [] Engine pitch follows speed in real time with per-class tone. - [] Drift squeal, tier ticks, boosts by tier, hop, land, bumps, scrape, crash all present. - [] Every item has pickup, use, and hit sounds. - [] Dynamic music: stems react to position and final lap. - [] Announcer: countdown, final lap, position callouts. Barks have cooldowns. - [] Mixer: ducking works; no clipping; peak below -1 dBFS.

Technical - [] 0 console errors; 60 s soak on every track with no stalls. - [] First load under 8 s on a good phone; per-track streaming. - [] 4-player lobby works end-to-end (code, join, ready, start). - [] Save/load:

coins, unlocks, garage builds persist.

3. Engine and backend recommendation (production)

Layer	Recommendation	Why
Engine	Unity 6 LTS, URP, C# (alternative: Unreal 5.5 mobile renderer if the team is Unreal-heavy)	Best mobile tooling, Addressables for streaming 100 characters × 20 tracks, big hiring pool, mature profiler (Frame Debugger, Memory Profiler), Burst/Jobs for kart physics and AI
Physics	Custom kinematic arcade kart controller (no wheel colliders), fixed 60 Hz, Burst-compiled; use Unity Physics only for static queries	Determinism-lite and cost
Netcode	Custom UDP via Unity Transport; host-authoritative items, client prediction for own kart; optional dedicated headless Linux servers	Fine-grained control over lag-tolerant items
Backend	Nakama (open source, on a managed cluster) for accounts, rooms, matchmaker, leaderboards, cloud saves; or PlayFab if the studio prefers a managed service	Rooms with short codes, leaderboards, storage
Server hosting	Edgegap / Hathora / AWS GameLift for dedicated relays; Cloudflare for static content	Low cost at low concurrency
Content delivery	Addressables on CDN, per-track and per-character bundles	Meets the <8 s load budget
Audio	FMOD Studio (adaptive layers, snapshots, ducking)	The dynamic score and RPM-driven engines need parameter-driven mixing
Analytics	Opt-in anonymous (Unity Analytics or self-hosted PostHog)	Funnel and performance

This repository's browser slice uses three.js, WebAudio and WebRTC (with a free signalling relay) so the design can be played now on any phone.

4. Milestones for the slice (8 weeks, team of ~25)

Week	Milestone
1	Kart controller prototype, camera, test track, touch + gamepad input
2	Drift, boost tiers, start boost, off-road, bump rules
3	Track pipeline (spline → mesh → AI line → minimap) + Track 1
4	Items (8), item boxes, HUD, CPU racers with rubber-banding
5	Garage (stats, preview, saves), characters (8), Tracks 2 to 4
6	Audio vertical slice: engine layers, drift, boost, music stems, announcer, mixer
7	Netcode: codes, lobbies, prediction, items over network, host migration
8	Polish, perf pass (30/60 fps), quality-bar sign-off, playtest

5. Slice KPIs

- 60 fps on iPhone 13+ and Pixel 8 class, 30 fps on iPhone XR / Pixel 5 class.
- Race start to first interaction: under 10 s.

- 90% of playtesters can explain all 8 items after one race.
- Net: 4 humans, 150 ms RTT, no teleporting karts.

Production Order, Risks, Performance Budget

1. Production order (full game, 24 months, ~85 people, \$10M)

Phase	Months	Output	Gate
Pre-production	0 to 4	Pillars, kart controller prototype, art pipeline, audio pipeline, netcode spike	Playable greybox kart on one track; "fun in 60 seconds"
Vertical slice	4 to 8	The slice (8 characters, 6 karts, 4 tracks, 4-player lobbies, items, drift, garage, full audio slice)	Quality-bar checklist all green; investors/internal greenlight
Systems production	8 to 14	Roster pipeline (templated rig, 10 sets), kart modular system (40+ bodies, parts), track pipeline (20 tracks), CPU AI, progression, cloud saves, daily challenge, time trial ghosts	8 tracks + 40 characters complete
Content production	14 to 20	Remaining 12 tracks, 60 characters, remaining bodies and parts, music (21 pieces), VO	All tracks playable; content lock
Hardening	20 to 24	Balance, perf, QA, localisation (14 languages), certification (Apple, Google), live-ops tooling	Release candidate
Post-launch	24+	Battle mode, seasons, extra tracks	Live

Ordering rules

1. Prove the feel before scaling content.
2. Build the tools (track spline pipeline, character template rig, kart modular assembly) before the content.
3. Audio runs in parallel from day one: engine and mixer prototypes go in with the kart controller.
4. Netcode spike in pre-production: the architecture must be proven with 4 humans + 8 CPUs before any track is final.
5. Streaming and memory budgets are enforced from the first track.

Team (peak)

Design 8, Engineering 22, Art (characters 10, karts 8, environment 14, VFX 4, UI 5), Animation 6, Audio 6 (sound designers 3, composers 2, implementer 1), QA 10, Production 4, Online/backend 5, Live-ops 3.

2. Risk list

#	Risk	Likelihood	Impact	Mitigation	Owner
1	Kart handling not fun / not readable	M	H	Prototype first, weekly blind playtests, tuneable physics JSON, assist options	Design
2	Netcode: items feel unfair at 150 ms	H	H	Victim-authoritative hits, fast-forward projectile, 120 ms rewind validation, early 4-human tests	Online
3	100 characters × animations scope	H	H	Templated rig; shared animation set with per-character poses (idle/victory); 10 art sets of 10; outsourcing plan	Art

#	Risk	Likelihood	Impact	Mitigation	Owner
4	Memory on mid-range phones	H	H	Addressables streaming; one track + 12 racers resident; texture budgets; ASTC	Tech art
5	Load time over 8 s	M	H	Core under 40 MB; async loading; splash-to-menu under 4 s	Engineering
6	Audio: too many voices, clipping, harshness	M	H	Voice limits, priority, master limiter, loudness QA	Audio
7	Rubber-banding perceived as unfair	M	M	Light bands; item balancing; telemetry	Design
8	Bluetooth gamepad variance	M	M	SDL-style mapping DB, test matrix	QA
9	Thermal throttling on long sessions	M	M	Adaptive quality, 30 fps fallback, particle caps	Engineering
10	Platform review (child safety, loot boxes)	L	H	No loot boxes; cosmetic-only; child-safe design	Production
11	Backend cost spikes	L	M	Serverless TTL rooms; P2P by default; relay only when needed	Online
12	IP safety (no resemblance to existing characters)	M	H	Original design, legal clearance review of all 100 names and silhouettes	Legal
13	Scope creep: Battle mode	M	M	Optional, post-launch	Production
14	Content balance (100 characters)	M	M	Class modifiers only (no unique stats); stat budget invariant	Design

3. Performance budget

3.1 Targets by tier

Tier	Devices	Frame rate	Resolution scale	Shadows	Racers
High	iPhone 14+, Pixel 8+, Galaxy S23+	60 fps	100% (up to 1170p)	Player + nearest 3	12
Mid	iPhone XR to 12, Pixel 5 to 7	30 fps stable (60 on menus)	75%	Player only	12 (reduced FX on far karts)
Low	older	30 fps	60%	Blob shadows	8

3.2 Per-frame budget (60 fps = 16.6 ms)

System	Budget
Game logic + physics + AI	3.0 ms
Render CPU (draw submission)	4.0 ms
GPU	12.0 ms
Audio mixing	1.0 ms
Network	0.5 ms
Headroom	2.0 ms

3.3 Rendering budget

- Draw calls ≤ 250 (≤ 150 on Mid). Triangles ≤ 350k visible (kart 12k LOD0, 6k LOD1, 2.5k LOD2).
- Materials: ≤ 40 unique per scene, shared atlases; GPU instancing for scenery.
- Particles ≤ 600 live; overdraw capped.
- Textures: ASTC 6×6 on mobile; kart 1024², character 512² + 1024² face; track atlases 2048².

3.4 Memory budget (on device)

Item	Budget
Total app memory (high)	≤ 1.8 GB; (mid) ≤ 1.1 GB
Track resident	≤ 220 MB
12 racers (kart + character)	≤ 160 MB
Audio resident	≤ 90 MB (music streamed from disk)
UI + fonts	≤ 40 MB

3.5 Download and load

- Initial app download ≤ 250 MB (stores allow more; keep low for conversion).
- Core boot ≤ 4 s on a good phone; menu to race ≤ 8 s for a cached track.
- Asset streaming: one Addressable bundle per track (≈ 60 MB), per character (≈ 8 MB), per kart body (≈ 4 MB), per music set (≈ 14 MB).
- Cache eviction: keep the last 3 tracks and 24 characters.

3.6 Network budget

- ≤ 25 KB/s per client; ≤ 60 KB/s for the host with 4 humans + 8 CPUs.
- 20 Hz snapshots, 60 Hz simulation; 100 ms interpolation buffer.

3.7 Battery and thermals

- 30 min of play $\leq 18\%$ battery on a mid phone; adaptive resolution scaling when thermal state is "serious".

3.8 Web slice budget (what this repo's build enforces)

- Core bundle (JS + essential audio) ≤ 6 MB gzipped; first interactive under 8 s on a good phone on a typical 4G connection.
- Adaptive pixel ratio (1.0 to 0.6) from frame time; 60 fps target, 30 fps fallback with the fixed 60 Hz simulation.